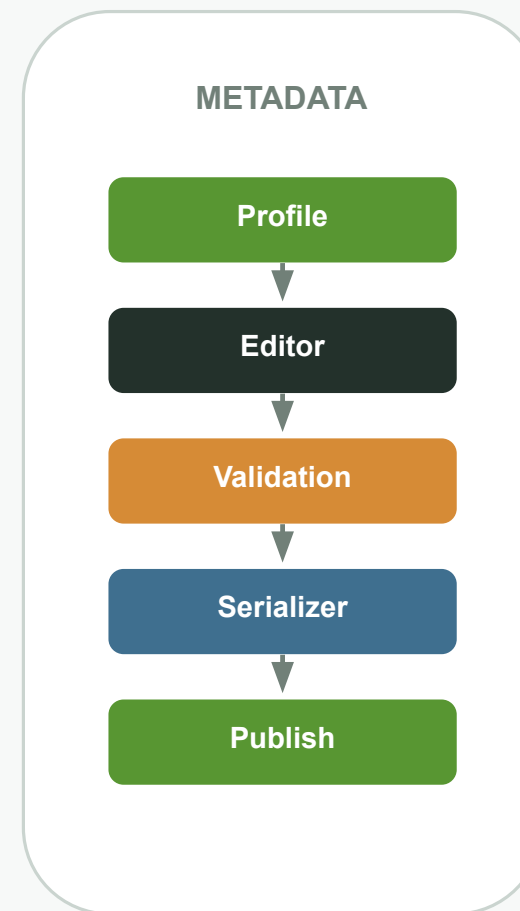


QGIS Metadata Profiles Framework

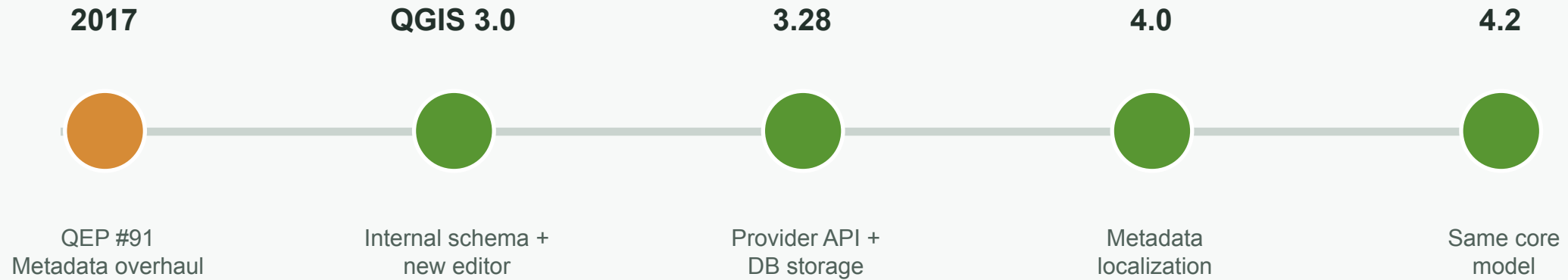
From an extensible metadata editor to a standards-aware framework



Discussion proposal • October 2026

How we got here

The current architecture is the result of a deliberate design choice.



The key 2017 decision

Use a QGIS internal metadata schema, independent from ISO or other external standards.

The consequence

External metadata must be mapped into the QGIS model — and some information may not have a place to go.

QGIS 4.x: strong storage, fixed model

The gap is not persistence. The gap is extensibility across profiles and standards.

What works well

- QMD sidecar metadata
- PostgreSQL storage
- GeoPackage storage
- Local SQLite catalog
- Browser / Metadata Search
- Provider-based storage

What remains constrained

- Fixed conceptual model
- Fixed editor structure
- Partial standard mappings
- Risk of lossy conversion
- Limited multilingual model
- Publishing is not a first-class layer

Core observation

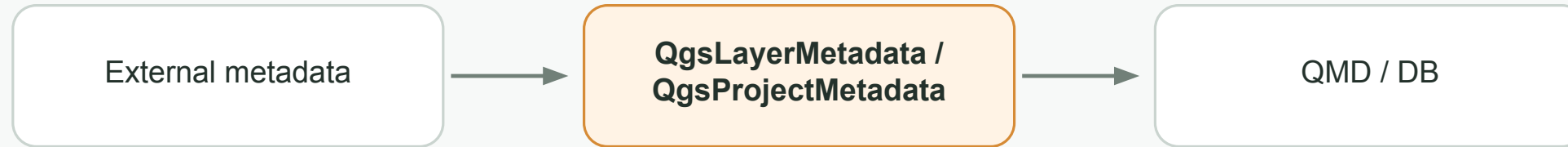
QgsAbstractMetadataBase is still an internal QGIS format. It is not a superset of every external metadata schema.

QGIS has a flexible metadata storage infrastructure, but a relatively fixed metadata model and editor.

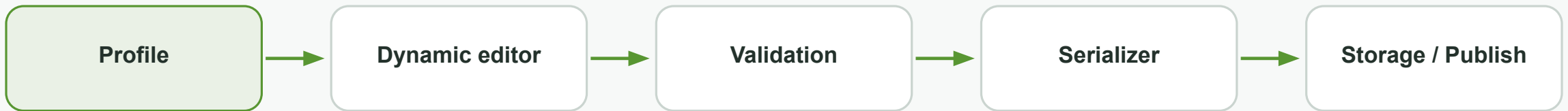
The problem to solve

We can store and retrieve metadata — but defining and exchanging profiles is harder.

TODAY



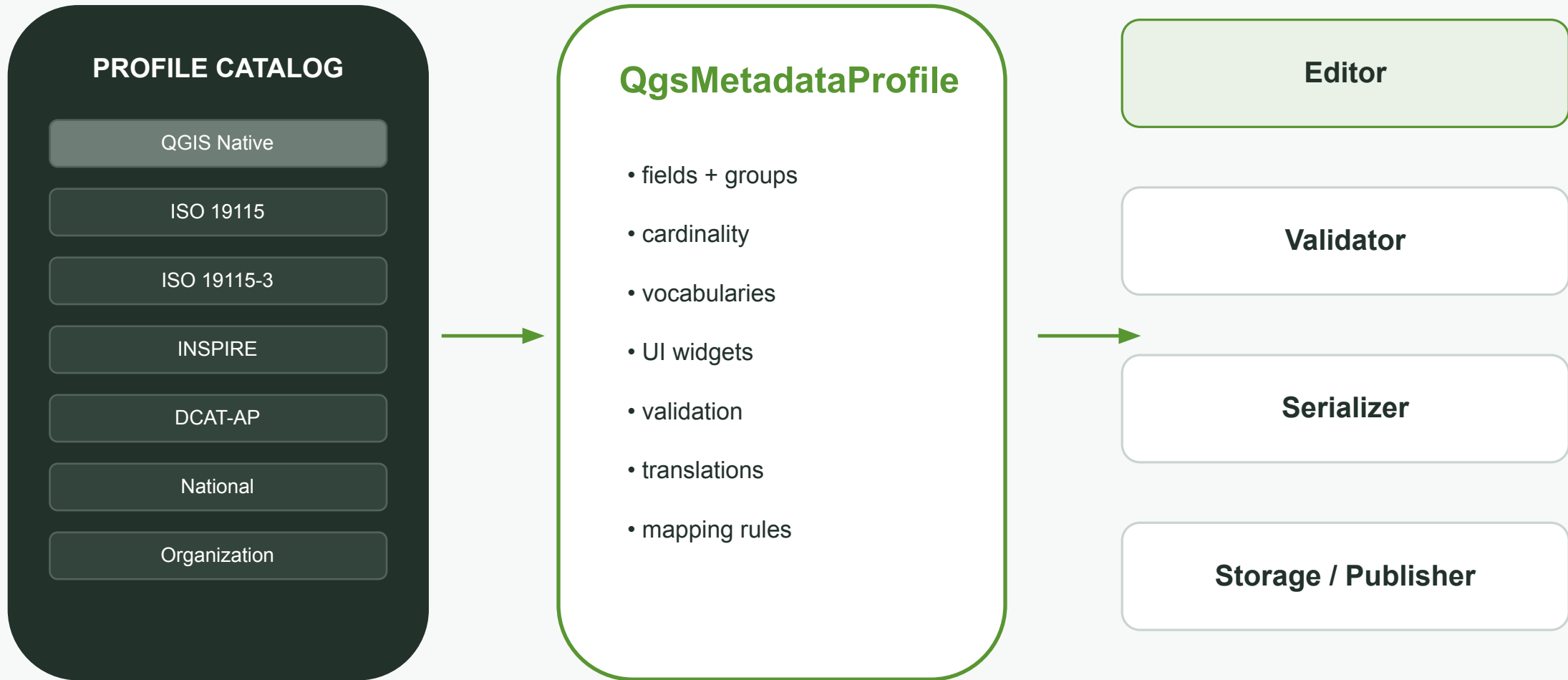
TARGET



Define → Edit → Validate → Transform → Exchange

Proposal: QgsMetadataProfile

Make the current QGIS metadata editor one profile among many.



Dynamic metadata editor

A profile definition drives both structure and user experience.

metadata-profile.json

```
{
  "profile": "ISO 19115-1 / Spain",
  "groups": [...],
  "fields": [...],
  "cardinality": {...},
  "vocabularies": {...},
  "validation": {...},
  "translations": {...},
  "mapping": {...}
}
```



Metadata — ISO 19115-1 / Spain

Identification

Title *

Abstract *

Distribution

Alternate title

Quality

Purpose

Status

Responsible parties

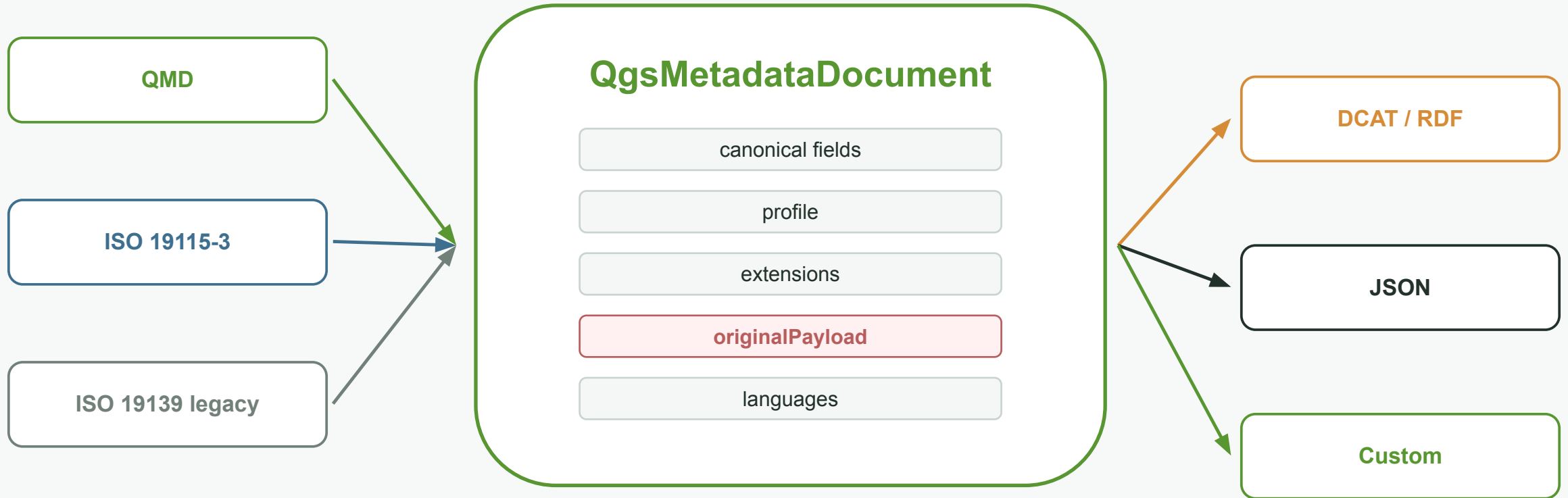
Language

Validate profile

GeoNode 5 provides a useful precedent: JSON Schema → validation + dynamic UI.

Serialization without losing the original

Separate the canonical working model from profile-specific payloads.



Round-trip principle: preserve what QGIS cannot represent canonically.

Multilingual metadata as data, not UI translation

Language variants should belong to the metadata object itself.

Current direction

QGIS 4.0 introduced localization of selected metadata values through project translation. This is useful, but it is not the same as a first-class multilingual metadata model.

Title

ES

EN

FR

es

Red hidrográfica

en

Hydrographic network

fr

Réseau hydrographique

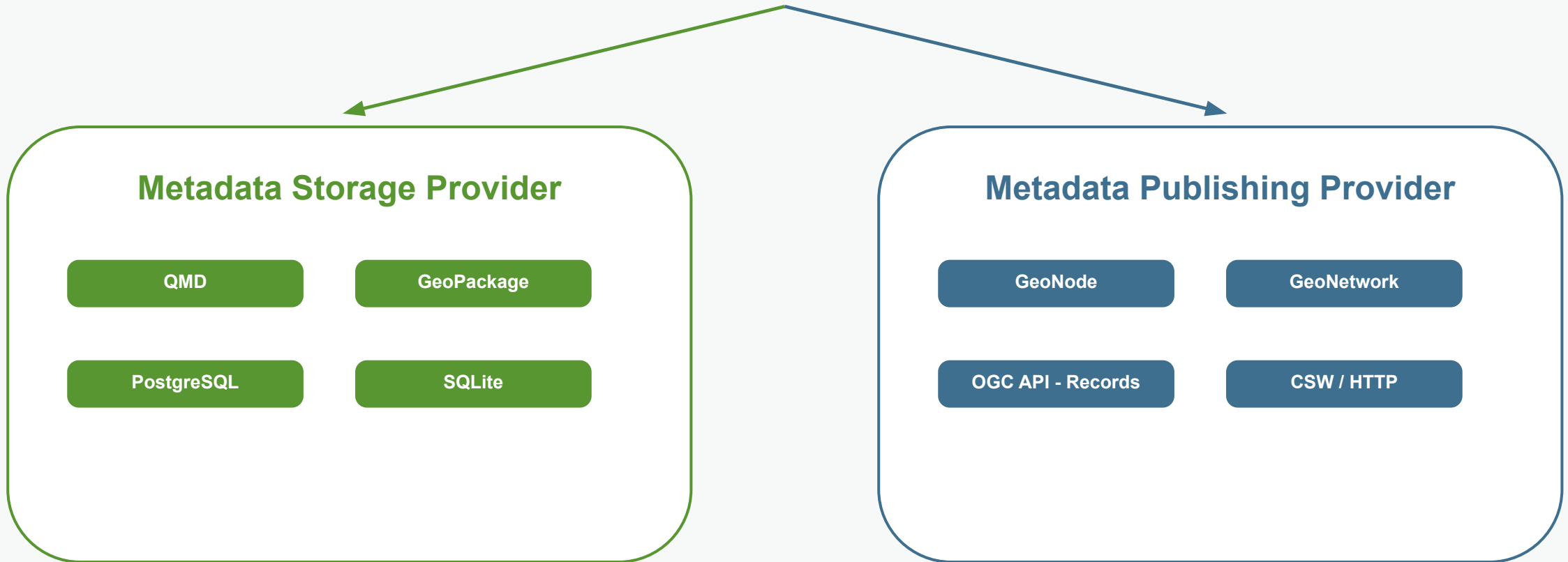
```
metadata.title["es"]    metadata.title["en"]  
metadata.title["fr"]
```

Useful for INSPIRE • UN • national catalogs • GeoNode • GeoNetwork

Separate storage from publishing

How metadata is described should not determine where it is stored or published.

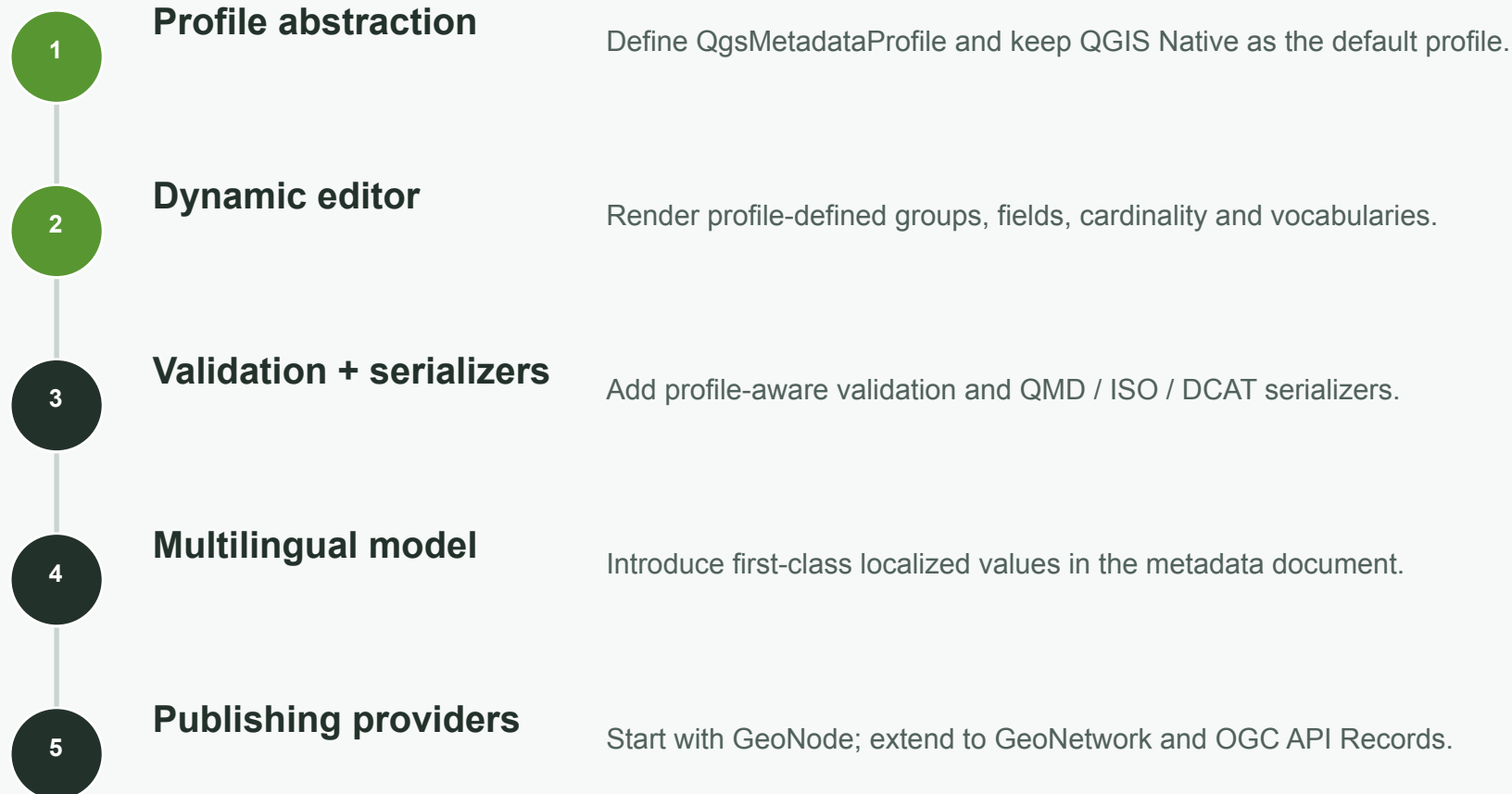
Metadata Document



Reuse the existing provider foundation; add publishing as a distinct extension point.

A pragmatic implementation path

Introduce the framework incrementally and preserve existing QGIS behavior.



Compatibility principle: existing projects and QMD remain valid throughout the migration.

The proposal in one sentence

**“I don’t think QGIS needs another metadata editor.
I think QGIS needs a metadata framework where the current QGIS
metadata editor becomes just one profile.”**

Dynamic Profiles

Standard Serialization

Multilingual Metadata

Pluggable Storage / Publishing

Key references

- QGIS QEP #91 — Overhaul Metadata Management in QGIS (2017)
- QGIS QEP #250 — Database Metadata Storage (2022)
- QGIS API — QgsAbstractMetadataBase
- GeoNode 5 — Metadata Management Architecture
- ISO 19115-1 and ISO 19115-3:2023
- GeoCat Bridge and PgMetadata plugins as existing ecosystem examples